Mechanistic Analysis of Over-Squashing and Attention Dilution in Hybrid GNNs

Problem, data, replications, pipeline, measures and first results

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MS Applied Data Science capstone

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Outline

1. Problem overview
2. Datasets
3. Replicating over-squashing and attention dilution
4. The pipeline and models
5. Measures
6. Results
7. Future work

Over-squashing: Jacobian decay and resistance bottlenecks

- Bottlenecks compress information from many nodes (Alon & Yahav 2021; Topping et al. 2022)
- Measured by the sensitivity of a node's output to a distant input:

$$J(u, v) = \left\| \frac{\partial h_v^{(L)}}{\partial h_u^{(0)}} \right\|_F \quad \text{decays with distance}$$

- Depends on width, depth and topology (Di Giovanni et al. 2023)
- Structural predictor: **effective resistance** $R(u, v)$; total resistance bounds over-squashing (Black et al. 2023)
- On trees $R(u, v)$ equals the hop distance

Attention dilution: entropy $\to \log N$, sensitivity $\sim 1/N$

- Global softmax attention: row $a_{i\cdot}$ is a distribution over N nodes
- Uninformative attention is uniform: entropy $H = \log N$, so each source has weight $1/N$
- Diagnostic: per-head entropy normalised by $\log N$

$$\tilde{H} = H/\log N \in [0,1], \qquad \tilde{H} \approx 1 \;\Rightarrow\; \text{attention fails to concentrate}$$

- Dilution in a *trained* model: $\tilde{H}$ stays near 1 as N grows
- Raw H rising with N is automatic and is not, by itself, evidence of dilution

The hybrid idea: an α-blend per layer

$$x' = x + \mathrm{Dropout}\big((1-\alpha)\, h_{\mathrm{loc}} + \alpha\, h_{\mathrm{glb}}\big)$$

- h_{loc}: GINE message passing; h_{glb}: multi-head self-attention
- $\alpha = 0$: pure MPNN $\alpha = 1$: pure transformer in between: GraphGPS-like (Rampášek et al. 2022)
- α is a fixed coefficient in the layer equation
- It is not "the share of information carried by attention"



Research question and hypotheses

Question. Do graph structure (effective resistance) and task range predict where on the local-to-global spectrum (best α) a model should sit? Can mechanistic measures explain why?

H1 At $\alpha = 0$, Jacobian sensitivity decays with $R(u, v)$ (negative log-log slope) more steeply than for the transformer; hybrids in between

H2 Normalised attention entropy rises with graph size for attention models

H3 Peptides-func is short-range (Bamberger et al. 2025), so expect a low best α

H4 Best α increases with effective resistance and with task range

Deliverable: a phase diagram. x = effective resistance, y = task range, colour = best α. The transformer's Jacobian slope serves as a size-confound baseline.

Gates: checks before any claim

Gate	Criterion	Status
G0	Rebuilt 4-layer hybrid, $\alpha = 0.5$, seed 0: test AP ≥ 0.652 on Peptides-func. $\alpha = 0$ and $\alpha = 1$ must also train	pending (Colab)
G1	Training is sound: mean test AP $\alpha = 0.5 \geq 0.63$, $\alpha = 1 \geq 0.60$, GCN ≥ 0.66 (or the gap documented)	pending
G2	Measurements are interpretable: Jacobian slope difference vs $\alpha = 1$ with a graph-clustered bootstrap CI	pending

If G1 fails, change one thing at a time on seed 0: training curves, then lr/warmup, then dropout, then BatchNorm vs LayerNorm, then 300 epochs.

Datasets at a glance

Dataset	Graphs	Task	Metric	Role
Peptides-func (LRGB)	15,535 molecules, ∼151 nodes	10-label graph classification	macro AP	main benchmark
Peptides-struct (LRGB)	same graphs	11-target regression	MAE	task-range axis (planned)
Reddit hyperlinks (SNAP)	1 graph, 35,776 nodes	node regression	rel. MSE per hop band	parallel node track
LamaH-CE rivers	32 gauge trees, 5–35 nodes	discharge, 1/3/7 days ahead	NRMSE, NSE	real-world tree test
Synthetic trees	96 trees, 63/127 nodes	near/far query sum	MSE	controlled test
synthetic-tiny	8–30 nodes	Peptides schema	AP	unit tests, smoke runs

Peptides-func and Peptides-struct (LRGB)

- 15,535 graphs; average ≈151 nodes, ≈307 edges, diameter ≈57 (Dwivedi et al. 2022)
- Split ≈70/15/15 (10,873 / 2,331 / 2,331)
- Atoms: 9 categorical columns. Bonds: 3 columns. All *embedded*, not passed through Linear
- RWSE positional encoding, 20 steps: $\mathrm{pe}_v = [(D^{-1}A)^k_{vv}]_{k=1}^{20}$

- **func**: BCE loss, macro AP
- **struct**: L1 loss, MAE (Phase 4)
- Same graphs, different task: a test of whether range depends on the task
- Caveat: the graphs are near-trees, so $R \approx$ hops. "Resistance beats hops" cannot be shown here

Reddit, rivers and synthetic trees

Reddit (SNAP) (Kumar et al. 2018)

- Subreddits; 286,561 body hyperlinks, used undirected
- 86 averaged post features
- **Distance split** by BFS from an anchor: train 0–1 hops, val 2, test 3+
- **Random control**: same sizes

LamaH-CE rivers (Klingler et al. 2021)

- Upstream gauge trees, depth 1–8
- 27 development and 5 held-out outlets
- Inputs: weather, or weather + flow
- Splits by time: 1981–99 / 2000–05 / 2006–17

Synthetic trees (v1)

- $N \in \{63, 127\}$, depth $\in \{4, 8\}$
- Early, uniform or late growth profiles
- 4 nodes marked at 1 hop, 4 at 4 hops; the root query picks near or far
- Reference MSE: oracle 0, topology-blind 0.5

What has been verified so far

Experiment	Purpose	Status
R1 Pilot LRGB replication	GCN / GPS vs published numbers	done **[pilot]**, 1 seed
R2 Pilot mechanistic notebook	Jacobian vs R, entropy vs size	done, flawed setup
R3 Pilot Reddit	virtual node helps far nodes	done **[pilot]**, unfair baseline
Unit tests	analytic sanity of all measures	125 passed (CPU)
Demos 1–3	mechanisms on real model code	**[illustrative, untrained]**
Synthetic tree sweep v1	MP vs hybrid vs attention	done, confounded
G0 / Phase 2 sweep / real Reddit	re-baseline and measure	not run

Pilot LRGB replication on Peptides-func [pilot]

Model	Val AP	Test AP	Published
GPS	0.6702	**0.6518**	0.6535
GCN	0.6545	**0.6263**	≈0.686

- RWSE, 4 layers, hidden 96, 1 seed
- GPS reproduces the published value
- GCN is 0.06 short of Tönshoff et al.'s tuned value, so this is only a partial replication
- G0's 0.652 threshold comes from this GPS run

Pilot mechanistic notebook [pilot]

Jacobian vs resistance (log-log slope, 240 pairs)

	GINE	Transf.	GPS
slope	-1.76	-0.48	-0.81
r	-0.78	-0.41	-0.69
test AP	0.54	0.27	0.24

Entropy vs size (400 graphs)

- Raw H rises for both models ($r = +0.88$, $+0.91$)
- $H/\log N$ rises *only for GPS* ($r = +0.60$), not for the transformer ($r = -0.09$)

Caveats: not citable as results

- 30 epochs, 1 seed; the attention models did not train
- Atom codes passed through `nn.Linear`; no positional encoding
- Entropy computed on head-averaged attention
- Shared LayerNorm; sum pooling

Nine bugs were fixed in Phase 1 (`gnn_mech`)

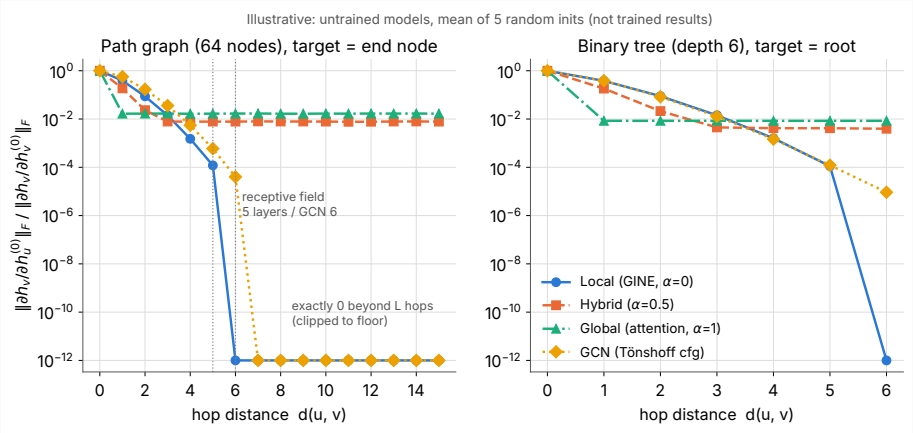
Pilot Reddit: virtual node vs SAGE by hop band [pilot]

Hop	n	SAGE	VMN	Cut
1	1,454	16,808	14,808	12%
2	17,876	90,056	56,115	38%
3	7,269	194,509	95,359	51%
4	613	207,113	77,806	62%
≥ 5	57	326,334	62,993	81%

Mean test MSE over 100 seeds (raw units)

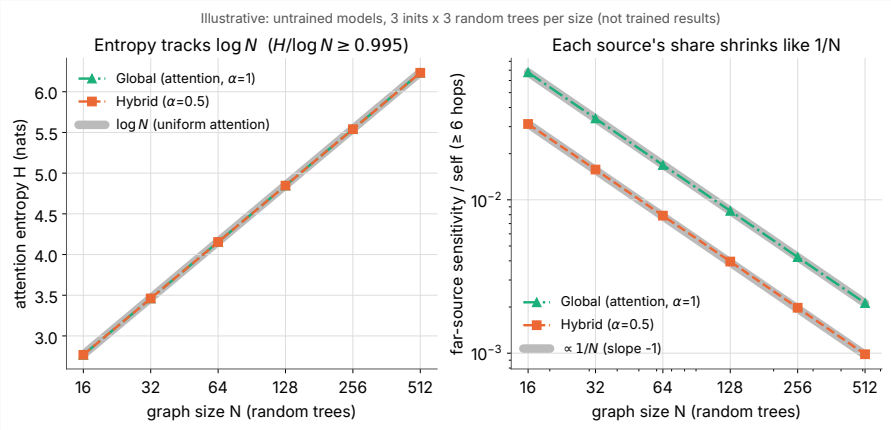
- The gain from a global shortcut grows with distance
- Caveats
  - Only VMN had an input projection (unfair baseline)
  - The target (character count) is almost a function of the node's own features
  - The distance split confounds "far" with "out of distribution"
- Phase 2 fixes all three

Demo 1: Jacobian decays with hops [illustrative, untrained]



MP loses ≈3–10× per hop and is exactly 0 beyond L hops. Attention is flat at $\approx 1/N$. The $\alpha = 0.5$ hybrid follows the local curve for 2–3 hops, then settles on a floor half as high.

Demo 2: attention dilution with graph size [illustrative, untrained]



Entropy lies on $\log N$ ($H/\log N \geq 0.995$). Sensitivity to sources $\geq$6 hops away falls exactly as $1/N$, e.g. global: 6.8×10^{-2} at $N = 16$ and 2.1×10^{-3} at $N = 512$.

Demo 3: range-measure sanity on a 150-node path [illustrative, untrained]

Model (untrained)	Node-level range, hops (mean / max)	Graph-level Hessian range
Local $\alpha = 0$ (5 layers)	0.67 / 0.69 (≤ 5 ✓)	undefined (ReLU-only)
Hybrid $\alpha = 0.5$	18.7 / 25.8	not run
Global $\alpha = 1$	34.3 / 47.0	not run
GCN (6 layers, GELU)	0.89 / 0.89	17.7

- The node-level range respects the receptive field and grows once attention is mixed in
- The Hessian range reaches ≈18 hops, far past GCN's 6-hop reach: mean pooling plus a nonlinear head couples all pairs

$\Rightarrow$ Models are compared using the **node-level** range

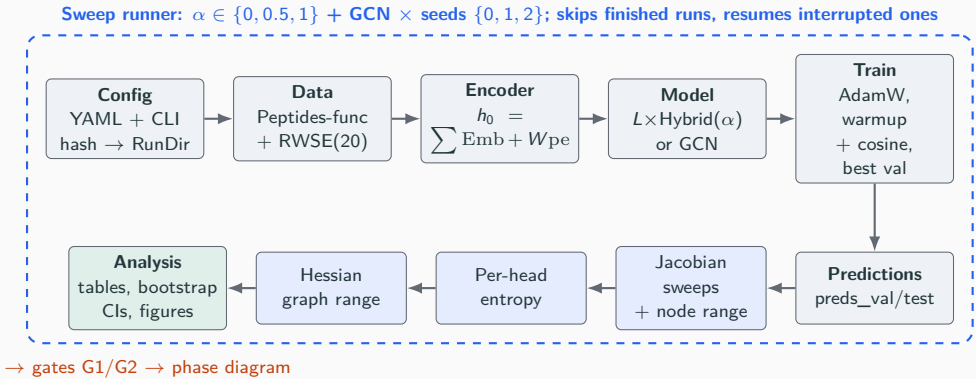
Synthetic tree sweep v1: MP fails the 4-hop query

Model	Near	Far	Balanced
MP ($\alpha = 0$)	0.0036	1.0094	0.5065
$\alpha = 0.25$	**0.0024**	0.0129	**0.0077**
$\alpha = 0.50$	0.0074	0.0147	0.0110
$\alpha = 0.75$	0.0050	**0.0122**	0.0086
Attn ($\alpha = 1$)	0.0093	0.0189	0.0141

Test MSE; predicting zero gives 1. 21 fits, about 49 min.

- MP solves the 1-hop query but fails the 4-hop one, though 4 hops lie inside its 8-layer reach
- Every hybrid beats pure attention
- Confounded: different target draws, mean aggregation for a sum target, attention-only hop bias, LayerNorm, unequal parameter counts
- Not re-verified here (archive missing)

End-to-end pipeline (gnn_mech)



Outputs per run: `metrics.json`, `measurements.json`, `best.pt`, `history.json`. All measures run on the best-validation checkpoint in eval mode.

HybridLayer, exactly as implemented

$$h_{\text{loc}} = \mathrm{BN}_{\text{loc}}\big(\mathrm{GINE}(x, E, e)\big), \qquad h_{\text{glb}} = \mathrm{BN}_{\text{glb}}\big(\mathrm{MHA}(x)\big)$$
$$x' = x + \mathrm{Dropout}\big((1-\alpha)\, h_{\text{loc}} + \alpha\, h_{\text{glb}}\big)$$
$$x'' = \mathrm{BN}_{\text{ffn}}\big(x' + \mathrm{FFN}(x')\big), \qquad \mathrm{FFN} = \mathrm{Lin}(d \to 2d)\ \mathrm{ReLU}\ \mathrm{Dropout}\ \mathrm{Lin}(2d \to d)$$

GINE branch

$$\mathrm{GINE}(x)_v = \mathrm{MLP}\Big((1+\epsilon)x_v + \sum_{u \in \mathcal{N}(v)} \mathrm{ReLU}(x_u + e_{uv})\Big)$$

Learnable ϵ; edge embeddings already have width d

MHA branch

- Dense attention *within* each graph, with a padding mask
- 4 heads, $\mathrm{softmax}(QK^\top/\sqrt{d_h})V$, attention dropout 0.1
- $\alpha = 0$: MHA not built; $\alpha = 1$: GINE not built

Model: InputEncoder → $L\times$ HybridLayer → mean pool → MLP head → 10 logits. All norms are BatchNorm, as in GraphGPS.

Baselines: GCN (Tönshoff-aligned) and the Reddit models

GCN (Tönshoff et al. 2023)

$$x \leftarrow x + \mathrm{Dropout}\big(\mathrm{GELU}(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}xW)\big)$$

- 6 layers, hidden 235, no per-layer norm, 3-layer head
- 250 epochs, batch 200, RWSE 20; ignores edge features
- Target ≈0.686 AP; remaining difference: how RWSE is injected

Reddit node regression

- **SAGE**: Linear(in→64) → 3×SAGEConv

$$h'_v = W_1 h_v + W_2 \,\mathrm{mean}_{u\in\mathcal{N}(v)} h_u$$

- **VMN**: the same network plus one master node linked to every node, so all pairs are within 2 hops
- VMN is the node-task analogue of $\alpha \to 1$
- Fix: both models now have the input projection; 30 seeds each

Model sizes and the α grid

Config	hidden	L	heads	params
Hybrid $\alpha = 0$ (GINE)	96	5	4	311,055
Hybrid $\alpha = 0.5$ (GPS-like)	96	5	4	498,255
Hybrid $\alpha = 1$ (transformer)	96	5	4	404,170
GCN (Tönshoff)	235	6	–	491,865
G0 pilot-equivalence $\alpha = 0.5$	96	4	4	404,654

LRGB budget is 500k. Training: 200 epochs, batch 128, AdamW lr 10^{-3}, 5 warmup epochs.

- **Phase 2**: $\alpha \in \{0, 0.5, 1\}$ + GCN, 3 seeds (12 runs)
- **Phase 3**: $\alpha \in \{0, .25, .5, .75, 1\}$, 3 seeds
- **Rivers**: $\alpha \in \{0, .25, .5, .75, 1\}$
- Confound: parameter count changes with α because unused branches are not built

Effective resistance and Jacobian sensitivity

Effective resistance

$$R(u, v) = (e_u - e_v)^\top L^+ (e_u - e_v)$$

- $L = D - A$; $L^+ = (L + \frac{1}{n} J)^{-1}$ on the largest connected component
- Graph level: mean of R over all pairs
- Tests: path $\Rightarrow R =$ hops; complete graph $\Rightarrow R = 2/n$

Jacobian sensitivity

$$J(u, v) = \left\| \frac{\partial h_v^{(L)}}{\partial h_u^{(0)}} \right\|_F$$

- Taken w.r.t. the embedded input h_0
- 200 graphs $\times$ 4 targets $\times$ 32 sources, stratified by R
- Slope b: $\log_{10} J = a + b \log_{10} R$ (also vs hops; size-controlled with $+\beta_N \log_{10} n$)
- G2: $\Delta b = b_\alpha - b_{\alpha=1}$, **graph-clustered bootstrap** (resample graphs, 1000 reps)
- One batched backward per target: $\approx$30$\times$ speedup

Per-head attention entropy (dilution)

For layer ℓ, head h, graph with N nodes and attention a_{ij}:

$$H_{\ell,h} = \frac{1}{N}\sum_{i=1}^{N}\Big(-\sum_{j=1}^{N} a_{ij}^{(\ell,h)} \log a_{ij}^{(\ell,h)}\Big), \qquad \tilde{H}_{\ell,h} = \frac{H_{\ell,h}}{\log N} \in [0,1]$$

- Computed **per head**; averaging heads first inflates entropy (a pilot bug)
- Uniform attention $\Rightarrow \tilde{H} = 1$ (unit-tested)
- 500 graphs: Pearson $r(N, H)$, $r(N, \tilde{H})$, per-layer means, graph-bootstrap CIs
- Undefined for $\alpha = 0$

α as the interpolation knob

$\alpha = 0$: receptive field of L hops (over-squashing regime). $\alpha > 0$: every pair connected in one layer (dilution regime). Jacobian slope, entropy, range and AP are all reported *as functions of* α; best α per resistance bin $\rightarrow$ phase diagram.

Range measure (Bamberger et al., ICML 2025)

Implemented following Gutteridge's `range-measure` repository.

$$\text{node: } I_{vu} = \sum_{k,d} \left|\partial y_{v,k} / \partial x_{u,d}\right|$$

$$\text{graph: } I_{uw} = \sum_{k,d,d'} \left|\partial^2 y_k / \partial x_{u,d}\, \partial x_{w,d'}\right|$$

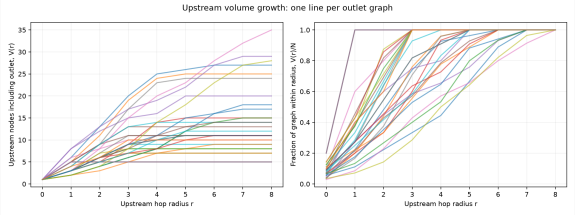
$$\rho_v = \frac{\sum_u I_{vu}\, d(v,u)}{\sum_u I_{vu}}, \quad d \in \{\text{hops}, R\}$$

Graph range is the mean over nodes; dataset range is the mean over graphs.

- **Node-level** (Jacobian): reuses the Jacobian sweeps; matches the paper's LRGB setting
- **Graph-level** (Hessian): 30 graphs, $\approx$50 s per graph on CPU
- Caveats: the Hessian is zero for ReLU-only $\alpha = 0$; pooling into a nonlinear head extends the range past the receptive field
- Reading rule: all hop ranges $\leq L/2 \Rightarrow$ short-range

River experiment: setup

- GRU temporal encoder (90 days) $\rightarrow$ 8 spatial layers, width 64, 4 heads
- Each layer: $(1-\alpha)\,\mathrm{LN}(\mathrm{MP}) + \alpha\,\mathrm{LN}(\mathrm{Attn})$
- MP: upstream$\rightarrow$downstream, mean aggregation
- Attention over ancestors, with a hop bias
- Baselines: outlet-only GRU, upstream pooling, persistence
- 72 fits, 3 seeds, $\approx$28.5 GPU-h
- Hypothesis: faster upstream growth $V(r)$ means a larger gain over MP



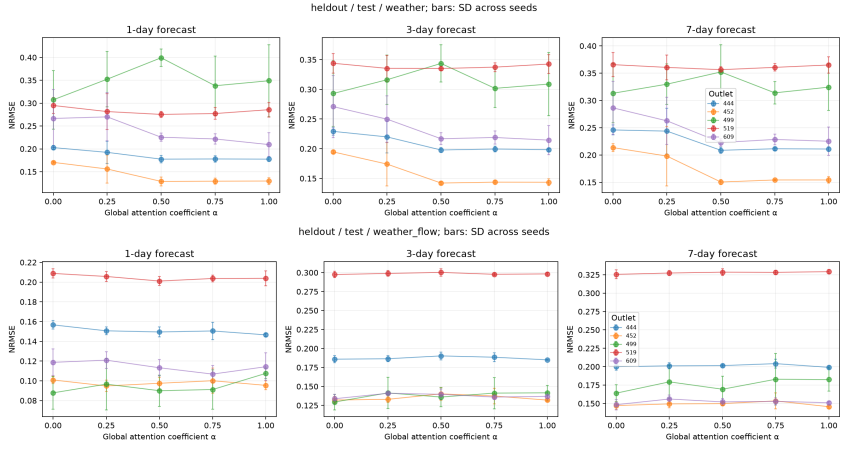
$V(r)$: upstream nodes within r hops. It saturates at N.

Held-out results: the α curve (test NRMSE, lower is better)

Model	Weather	Weather+flow
MP ($\alpha = 0$)	0.2666	0.1691
$\alpha = 0.25$	0.2628	0.1722
$\alpha = 0.5$	0.2489	0.1706
$\alpha = 0.75$	**0.2409**	0.1716
Attention ($\alpha = 1$)	0.2426	0.1713
Upstream pooling	0.2437	**0.1689**
Outlet-only GRU	0.2600	0.1696

- **Weather only**: $\alpha = 0.75$ is best, 9.6% better than MP
- But only 0.7% better than attention and 1.1% better than pooling, so the gain comes mostly from *global access*
- **Weather+flow**: no gain; every $\alpha > 0$ is slightly worse than MP and pooling is best
- Validation-chosen α is unstable once flow is given
- Mean NSE in the weather regime is negative (-0.30 at $\alpha = 0.75$)

α curves per held-out outlet



Top: weather (most outlets improve up to $\alpha \geq 0.5$; the largest graph, 499, does not). Bottom: weather+flow (flat).

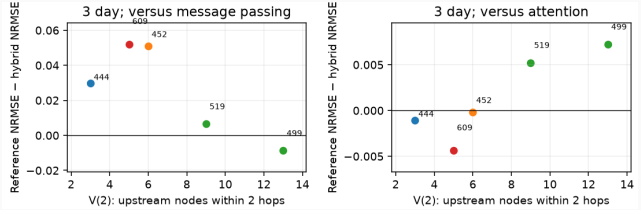
Context ablation: a hint of over-squashing

Weather	Full	Radius 2	Radius 0
MP ($\alpha = 0$)	0.2666	0.2673	**0.2584**
$\alpha = 0.75$	**0.2409**	0.2695	0.2741
Attention	**0.2426**	0.2583	0.2791

Held-out test NRMSE, mean over horizons. Radius r: dynamic inputs kept only within r hops.

- MP does *better* when upstream inputs are hidden (same on the development set)
- Attention and the hybrid exploit upstream context: about -12% NRMSE vs radius 0
- Consistent with over-squashing, but also with optimisation, mean-aggregation or parameter effects
- The ablation masks inputs; it does not rewire the graph

Growth → gain hypothesis: not supported

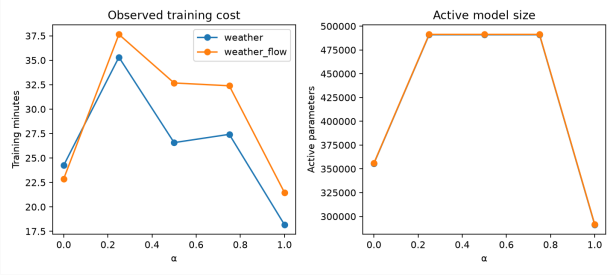


Held-out, weather, $\alpha = 0.75$, 3-day horizon. Positive = hybrid better than the reference.

Outlet	Nodes	$V(2)$	Gain vs MP
444	9	3	+13.1%
609	7	5	+18.8%
452	8	6	+26.2%
519	11	9	+2.9%
499	29	13	−4.4%

- The fastest-growing outlet (499) gains *least*
- Spearman $\rho(V(2), \text{gain})$ is ≈ -0.7 vs MP but $+0.9$ vs attention: **the sign flips with the reference**
- $n = 5$; leave-one-group-out ranges span both signs
- Development set: after adjusting for size, depth and area, $\rho \approx 0$
- A graph-conditioned α selector is 2.3% *worse* than a fixed α

Cost vs α



Model	Params	s/epoch
MP	355,715	55
Hybrid	490,979	80–82
Attention	291,299	55
Pooling	26,627	15
Outlet GRU	22,403	11

- A hybrid takes $\approx 1.5\times$ the time per epoch of either endpoint
- Pooling matches the best hybrid with $\approx 5\%$ of the parameters

Status: what does *not* exist yet

No Phase 2 real-data GPU results yet

- G0 not run, so the rebuilt hybrid has not been checked against 0.652 AP
- Phase 2 sweep (12 runs) not run: no re-baselined AP, no G1/G2, no Jacobian slope CIs, no entropy-vs-size on correctly built models, no range on real Peptides
- Aligned Tönshoff GCN not run; only the pilot's 0.6263 exists
- Reddit not run on the real graph (SNAP is blocked in the sandbox)

What is solid

Code for Phases 1 and 2 is built and CPU-tested (125 tests pass). Demos confirm the measures behave as theory predicts. River and tree results are real, but small or confounded.

Next: run on Colab (in order)

1. **G0**: `pilot_equivalence.yaml`, $\alpha = 0.5$ needs ≥ 0.652 AP; $\alpha = 0$ and $\alpha = 1$ must train
2. **Phase 2 sweep**: 12 runs, $\approx$5 GPU-h ($\approx$4 h training + $\approx$1 h measuring)
3. **Range sanity** on real Peptides: $\alpha = 0$ mean node range ≤ 5 hops
4. **Real Reddit**: reproduce the pilot's band ordering, select a target from the data, then 30 seeds $\times$ {SAGE, VMN} $\times$ {distance, random}
5. **Analysis notebook** $\rightarrow$ check G1/G2 $\rightarrow$ `docs/results/phase2.md`

Later phases

P3 Full α sweep $\{0, .25, .5, .75, 1\} \times 3$ seeds; best α per resistance bin; gated-attention baseline

P4 Task-range axis: Peptides-func vs Peptides-struct

P5 Synthetic suite: controlled resistance and long-range dependence, distractor injection $\rightarrow$ **phase diagram**

P6 Predictor of best α from structure + range; held-out check

Tree experiment v2 (planned, not run)

- Residual Sum-MP/Attention: sum aggregation scaled by $1/\sqrt{\Delta}$, no attention-only hop bias, no amplitude-stripping LayerNorm
- Addressed retrieval: $K = 4$ vs 16 candidates at 4 hops
- Parameter-matched endpoints
- A **dilution intervention** that adds irrelevant nodes

Takeaways

- The **mechanisms** show up in real model code: MP sensitivity falls by orders of magnitude within L hops, and attention sensitivity falls as $1/N$ (untrained demos)
- **Global access helps when the task needs distant information**: synthetic far query, weather-only rivers ($\alpha = 0.75$ is 9.6% better than MP)
- When local history is enough (weather+flow), MP or pooling suffices and the hybrid only adds cost
- Graph growth does not predict the hybrid's gain: the sign flips with the reference model
- The pilot Peptides evidence is flawed; the rebuilt, tested pipeline is ready for G0 and the Phase 2 sweep
- Goal: a phase diagram of best α against effective resistance and task range

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